

June-August 2026 arXiv math.DG SARE Ranking Report

The combined Top 100 is presented first, followed by the official monthly rankings for June, July, and August and the 10 August new/cross-list batch. S_H is the human-residual score after discounting work estimated to be AI-replaceable, and S_eq=(V+S_H)/2 is the ranking score. Main claims are assumed correct; no paper was checked line by line. Coverage: 788 official entries; 238 papers in the official monthly 30% pools, and 239 papers in the combined comparison set. Top 100 composition: June 37, July 49, August 14. Cutoff: 90.175.

Overall Ranking - Ranks 1-50

Rank	Month	arXiv	Pages	AI%	S_H	S_eq	Title / Authors	Assessment
1	2026-06	2606.01511	119	8	94.90	95.30	A Branch Set Stratification for Stationary Varifolds with Epsilon-Regularity Brian Krummel; Paul Minter; Neshan Wickramasekera	Long branch-set stratification and epsilon-regularity framework for stationary varifolds.
2	2026-06	2606.03583	58	8	94.90	95.07	Closed minimal surfaces of index one in Riemannian manifolds Fernando C. Marques; André Neves	Index-one minimal surface theory by leading min-max authors; high conceptual and technical weight.
3	2026-06	2606.12257	333	8	94.70	94.85	Quantum cohomology and split generation in Lagrangian Floer theory Mohammed Abouzaid; Kenji Fukaya; Yong-Geun Oh; Hiroshi Ohta; Kaoru Ono	Massive foundational Floer-theoretic contribution with strong downstream value.
4	2026-06	2606.30098	77	8	94.05	94.38	Min-Max Construction of Anisotropic Minimal Surfaces with Genus Bound Antonio De Rosa; Aria Halavati; Ling Wang	Min-max construction with genus control in anisotropic surface theory.
5	2026-07	2607.07134	116	8	94.00	94.35	Naked Singularities beyond Spherical Symmetry: Singular Inner Cauchy Horizons for the Einstein-Scalar Field System Xinliang An; Shengrong Wu	Major GR-PDE construction beyond spherical symmetry; strong geometric analysis relevance.
6	2026-07	2607.13356	156	8	94.00	94.35	Analysis of singularities of area-minimizing currents, Part III: branch points of planar frequency I= 2, higher order asymptotics, and the local topology Brian Krummel; Neshan Wickramasekera	Large-scale regularity/singularity analysis for area-minimizing currents; exceptional human proof complexity.
7	2026-06	2606.13335	59	8	93.75	94.07	Sharp stability of Alexandrov's theorem for C^1 domains in the small-excess regime Alessio Figalli; Yi Ru-Ya Zhang	Sharp quantitative stability for Alexandrov-type rigidity.
8	2026-06	2606.15795	44	8	93.75	94.07	Entire area-minimizing surfaces in R^n of density two are quadratic or planar Nick Edelen; Luis Atzin Franco Reyna; Paul Minter	Sharp classification for density-two entire area minimizers.
9	2026-06	2606.22355	-	8	93.75	94.00	Uncountably many non-rotationally symmetric type II ancient Yamabe flows on the sphere Haixia Chen; Seunghyeok Kim; Monica Musso	Constructs large family of type-II ancient Yamabe flows with symmetry breaking.
10	2026-07	2607.15522	34	9	93.00	93.65	Nonnegative Bakry–Émery Curvature on Bounded-Degree Graphs Implies Volume Doubling and Poincaré Inequalities Qi Guo; Xueping Huang; Yi C. Huang	Bounded-degree CD(0,∞) graphs are uniformly volume doubling and satisfy a scale-invariant L2 Poincaré inequality; settles the Cushing-Liu-Peyerimhoff polynomial-growth conjecture in a stronger form.
11	2026-06	2606.00964	56	9	93.40	93.55	Mean convex flows with surgery Gabor Székelyhidi	Mean-convex MCF with surgery; high technical geometric-flow relevance.
12	2026-06	2606.15816	-	9	93.40	93.55	Kähler-Ricci Flow: from divisors to cusps Eleonora Di Nezza; Vincent Guedj; Chinh H. Lu	Kähler-Ricci flow transition from divisorial to cusp singularities.
13	2026-08	2608.07240	59	10	92.90	93.45	Critical Geller Equations on Complex Hyperbolic Space: Nondegeneracy, Geodesic Symmetry, and Global Compactness Jüנגang Li	Builds a broad critical Geller/Brezis-Nirenberg theory: sharp constants and classification, invariant-sector profiles and stability, nondegeneracy, compactness, symmetry, asymptotics, and perturbative existence.
14	2026-06	2606.30130	74	9	93.05	93.38	Gromov's dihedral rigidity conjecture in dimension three Jinmin Wang; Zhizhang Xie; Guoliang Yu	Major scalar-curvature rigidity conjecture in dimension three.
15	2026-07	2607.03152	110	9	93.15	93.33	Finite-Time Singularities of Lagrangian Mean Curvature Flow with Quantitatively Precise Dynamics Maxwell Stolarski; Wei-Bo Su	Quantitative singularity dynamics for Lagrangian MCF; technically deep and reusable.
16	2026-07	2607.03383	78	9	93.15	93.33	Unique asymptotics of SO(k)×SO(n-k+1) symmetric ancient ovals of Ricci flow Panagiota Daskalopoulos; Wenkui Du; Natasa Sesum; Ziyi Zhao	Ancient Ricci-flow asymptotics by leading flow techniques; high geometric PDE value.
17	2026-07	2607.11236	-	9	93.15	93.33	Formation of Trapped Surfaces from Spacelike Initial Data Xuantao Chen; Sergiu Klainerman	High-value trapped-surface formation result; long nonlinear geometric PDE proof architecture.
18	2026-08	2608.05128	148	9	93.15	93.33	Large mass limits of G_2 and Calabi-Yau monopoles: calibrated concentration, Higgs zeros, and abelianization Daniel Fadel; Gonçalo Oliveira	Very large gauge/calibrated-geometry work; high human proof burden and structural value.
19	2026-06	2606.29135	27	9	93.05	93.15	Gromov's Simplicial Volume Vanishing Conjecture for Positive Scalar Curvature and Fundamental Group Decay Qiaochu Ma; Guoliang Yu	Important PSC-topology result toward Gromov-type vanishing conjectures.
20	2026-06	2606.21062	-	9	92.75	93.07	The prescribed Hermitian-Yang-Mills flow I Zhiyao Xiong; Xiaokui Yang; Shing-Tung Yau	First part of prescribed HYM flow program; strong complex differential geometry content.
21	2026-06	2606.21073	-	9	92.75	93.07	The prescribed Hermitian-Yang-Mills flow II Zhiyao Xiong; Xiaokui Yang; Shing-Tung Yau	Second part of prescribed HYM flow program; complements Part I.
22	2026-08	2608.06320	18	8	92.45	93.00	Dihedral Rigidity for Convex Polytopes by Smooth Approximation Yuchen Bi	Smooth-approximation proof of Gromov's dihedral rigidity conjecture for convex polytopes.
23	2026-08	2608.07231	25	10	92.50	92.72	Liouville Rigidity for Entire Strictly Spacelike Solutions of a Lorentzian Prescribed Mean Curvature Equation Xi-Nan Ma; Tian Wu; Wangzhe Wu; Bao Yu	Removes radial, decay, integrability, and uniform-gap assumptions in a subcritical Lorentzian prescribed-mean-curvature Liouville problem, using global Lorentz-factor bounds and new weighted tensor identities.
24	2026-06	2606.13301	-	9	92.55	92.62	Gromov-Hausdorff convergence of time-slices of singular Ricci flows in dimension three Yu Li	GH convergence analysis for singular 3D Ricci flows.
25	2026-06	2606.21541	64	9	92.55	92.62	Spectral Analysis for Finite-Time Singularities of Lagrangian Mean Curvature Flow Maxwell Stolarski; Wei-Bo Su	Spectral analysis around finite-time singularity models in Lagrangian MCF.
26	2026-08	2608.03575	39	10	92.10	92.50	A Gauss-Bonnet-Type Dichotomy for Unimodular Random Infinite Trivalent Hyperbolic Polyhedra Huabin Ge; Yangxiang Lu; Chuwen Wang; Tian Zhou	Unimodular Gauss-Bonnet dichotomy, finite-limit obstruction, a refined ring lemma, Poisson-boundary identification, and positive hyperbolic speed.
27	2026-07	2607.03930	-	10	92.30	92.45	De Giorgi varifold solutions to Mean Curvature Flow: a minimizing movements approach Tim Laux; Andrea Poiatti	Variational weak solution framework for MCF; high method transfer potential.
28	2026-07	2607.06688	-	10	92.30	92.45	Poisson K-stability and the semiclassical Yau-Tian-Donaldson correspondence Vestislav Apostolov; Brent Pym; Jeffrey Streets	Connects K-stability/YTD ideas with Poisson geometry; high conceptual breadth.
29	2026-07	2607.20741	-	10	92.30	92.45	The Penrose conjecture for initial data sets satisfying a 2-convexity condition Conghan Dong	Penrose-type result under geometric convexity; important in scalar-curvature/GR interface.
30	2026-07	2607.23879	40	10	92.30	92.45	Closed mean curvature flows with prescribed tangent flows Jingwen Chen; Tang-Kai Lee; Ao Sun; Xinrui Zhao	Prescribed tangent-flow phenomena for closed MCF; strong singularity-theory relevance.
31	2026-07	2607.25644	62	10	92.30	92.45	Gromov-Hausdorff Limits of Noncollapsed Kähler-Ricci Flows and the Geometry of Ricci Shrinkers Yu Li; Junsheng Zhang	Noncollapsed KRF limits and shrinker geometry; central geometric-flow problem area.
32	2026-07	2607.29502	-	10	92.30	92.45	A dichotomy for minimal submanifolds Tobias Holck Colding; William P. Minicozzi II	Conceptual dichotomy statement in minimal-submanifold theory by leading authors.
33	2026-07	2607.29632	60	10	92.30	92.45	Regularity of branched stable minimal immersed hypersurfaces Gaoming Wang; Xuwen Zhang	Regularity theory for branched stable minimal hypersurfaces; substantial proof burden.
34	2026-08	2608.01682	42	10	92.30	92.45	Positive mass theorems for singular asymptotically hyperbolic manifolds Yuguang Shi; Chengzhang Sun; Zijun Wang	Positive mass theorem in singular asymptotically hyperbolic setting; strong geometric analysis.
35	2026-06	2606.02396	22	10	92.05	92.38	Existence of free boundary minimal disks in convex regions Lorenzo Santaro; Douglas Stryker; Zhichao Wang; Xin Zhou	Existence result for free-boundary minimal disks in convex regions.
36	2026-06	2606.24682	-	10	92.05	92.38	The linearized minimal surfaces problem Haim Grebnev; Plamen Stefanov; Gunther Uhlmann; Hanming Zhou	Linearized inverse-type problem for minimal surfaces with strong analysis component.
37	2026-06	2606.24754	-	10	92.05	92.30	Existence of classical minimal surfaces in 4 and 5-manifolds Da Rong Cheng; Xin Zhou	Classical minimal surface existence in low-dimensional manifolds.
38	2026-06	2606.13011	71	10	91.75	92.08	Complete Calabi-Yau metrics on noncompact K3 fibered threefolds Ruiming Liang	Construction and analysis of complete CY metrics on noncompact K3-fibered threefolds.
39	2026-06	2606.16033	-	10	91.75	92.08	Admissibility, Topology, and the sigma_k-Loewner-Nirenberg problem Hao Fang; Matthew J. Gursky	Topology and admissibility in the sigma_k-Loewner-Nirenberg problem.
40	2026-06	2606.22441	22	10	91.75	92.08	Parabolic rectifiability of the Brakke flow Yu Tong Liu; Myles Workman	Parabolic rectifiability for Brakke flows; useful geometric measure theory tool.
41	2026-07	2607.27162	23	10	91.65	91.88	Pointwise subexponential growth and near-diffusive displacement on bounded-degree graphs with non-negative Ollivier–Ricci curvature Chiyu Zhou	Near-diffusive displacement and pointwise subexponential volume growth for bounded-degree graphs with nonnegative Ollivier-Ricci curvature.
42	2026-06	2606.04586	43	10	91.75	91.85	Calibration energy and mean curvature flow Tatsuya Miura; Fabian Rupp	Calibration-energy perspective for MCF; technically robust.
43	2026-06	2606.08006	23	12	91.55	91.78	Optimal spectral rigidity of the hypercube via Bakry–Emery curvature Yanlong Ding; Shiping Liu; Chiyu Zhou	Sharp discrete curvature/spectral rigidity on the hypercube.
44	2026-08	2608.07263	16	12	91.20	91.50	Diagonalization of the Toda flow for arbitrary isospectral symmetric matrices David Martínez Torres; Carlos Tomei	Constructs a dense covering atlas that linearizes the Toda flow on arbitrary symmetric isospectral manifolds, based on a new constrained QLU factorization of independent interest.
45	2026-06	2606.08086	28	11	91.05	91.38	Monotone quantities on 3-manifolds with nonnegative scalar curvature Jiannan Chen; Haiping Fu	Monotone quantities in 3D nonnegative scalar curvature geometry.
46	2026-06	2606.12063	-	11	90.75	91.07	Optimal geometric estimates for compact Kähler manifolds of a Nash entropy bound Weiqi Zhang; Yashan Zhang	Optimal estimates under Nash entropy bounds in compact Kähler setting.
47	2026-06	2606.12379	21	11	90.75	91.07	A Local Singularity Analysis for the Ricci Flow and its Applications to Ricci Flows with Bounded Scalar Curvature -- Part II Reto Buzano; Gianmichele Di Matteo	Local singularity analysis for scalar-curvature-bounded Ricci flows.
48	2026-06	2606.19801	-	11	90.75	91.07	Positive mass theorem and the Yamabe equation on CR manifolds Jih-Hsin Cheng	Positive mass theorem linked to CR Yamabe equation.
49	2026-06	2606.19806	10	11	90.75	91.07	The top Yau-Yang conjecture for Kähler manifolds with positive sectional curvature Ved V. Datar; Vamsi P. Pingali; Harish Seshadri	Concise result on the top Yau-Yang conjecture for positively curved Kähler manifolds.
50	2026-06	2606.19858	16	11	90.75	91.07	Spectral Positive Mass Theorem for Asymptotically Hyperbolic 3-manifolds with Toroidal Infinity Xiaoxiang Chai; Yimin Chen; Juncheol Pyo	Spectral positive mass theorem in AH three-manifolds with toroidal infinity.

June-August 2026 Overall Ranking - Ranks 51-100

The ordering uses S_eq, with V as the secondary tie-breaker and arXiv number as the display tie-breaker.

Rank	Month	arXiv	Pages	AI%	S_H	S_eq	Title / Authors	Assessment
51	2026-06	2606.28897	16	12	90.90	91.07	Scalar-Flatness for Critical Metrics of the L2-Scalar Curvature Functional in Dimensions $5 < n \leq 9$ Heng Zhang	Scalar-flatness for critical L2-scalar curvature metrics in middle dimensions.
52	2026-07	2607.02067	31	11	90.95	91.05	Higher regularity of the inverse anisotropic mean curvature flow Chaoqun Gao; Yong Wei; Rong Zhou	Regularity upgrade for inverse anisotropic MCF; solid geometric flow result.
53	2026-07	2607.02412	-	11	90.95	91.05	Cartan's and Gauss's equations and rigidity theorems for isometric embeddings in low Sobolev regularity Isaac Newell; Luc Nguyen	Low-Sobolev rigidity for isometric embeddings; high technical and analytic value.
54	2026-07	2607.06698	151	11	90.95	91.05	Infinity-harmonic functions and inverse mean curvature flow clusters Kai Xu	Very long analysis of infinity-harmonic and IMCF cluster structures.
55	2026-07	2607.08325	30	11	90.95	91.05	Geometric smoothing by the Kähler-Ricci Flow Eleonora Di Nezza; Vincent Guedj; Chinh Hoang Lu	KRF smoothing in complex geometric analysis; useful technical input.
56	2026-07	2607.08631	32	11	90.95	91.05	Existence of two embedded minimal spheres in S^3 with an arbitrary metric Zhichao Wang; Xin Zhou	Existence result for embedded minimal spheres; strong min-max/geometric analysis value.
57	2026-07	2607.12305	64	11	90.95	91.05	Capillary prisms, Coxeter gluing, and the π_2 -systole of 3-manifolds with positive scalar curvature Shihang He; Chao Li	Positive scalar curvature, capillary geometry and systolic estimates; broad geometric interest.
58	2026-07	2607.17459	88	11	90.95	91.05	The deformed Vortex equations and equivariant stability conditions Tristan C. Collins; Yukai Zhang	Gauge/Kähler stability problem with substantial length and technical structure.
59	2026-07	2607.18216	-	11	90.95	91.05	Sharp Weitzenböck and PIC2 Estimates from Sectional-Scalar Curvature Pinching Jian Ge	Sharp curvature estimates from mixed sectional-scalar pinching; strong comparison geometry.
60	2026-07	2607.18792	35	11	90.95	91.05	Existence of global $W^{\{2,p\}}$ -isometric immersions for negatively curved surface metrics with unbounded second fundamental form Siran Li	Global low-regularity isometric immersion under negative curvature; strong PDE-geometry link.
61	2026-07	2607.18989	103	11	90.95	91.05	Diffeological non-Abelian Hodge theory: relative harmonic metrics and deformation theory Mahmud Azam; Steven Rayan	Large deformation/Hodge theory work with differential-geometric analytic component.
62	2026-07	2607.19091	32	11	90.95	91.05	On the sharp constants in curl-Sobolev inequalities on S^n Guofang Wang; Mingwei Zhang	Sharp inequality work on spheres; relevant across DG, AP and spectral theory.
63	2026-07	2607.19320	-	11	90.95	91.05	A spacetime positive mass theorem with corners via mollification Hyun Chul Jang; Aidan F. Wood	Positive mass theorem with corners; important low-regularity GR geometry theme.
64	2026-07	2607.24281	45	11	90.95	91.05	Calabi surgery for $Z/2$ harmonic 1-forms Jiahuang Chen; Siqi He; Dashen Yan	Surgery construction for $Z/2$ harmonic 1-forms; relevant to gauge/geometric topology.
65	2026-07	2607.24735	-	11	90.95	91.05	Area-minimizing submanifolds are not generically smooth, except for geodesics, minimal surfaces, and minimal hypersurfaces Zhenhua Liu	Striking generic regularity/nonregularity statement for area minimizers.
66	2026-07	2607.25015	36	11	90.95	91.05	Topological Rigidity of Contractible 3-Manifolds and Handlebody Interiors under Nonnegative Scalar Curvature Jiangcheng You; Heng Zhang	Scalar-curvature rigidity for contractible 3-manifolds; strong geometric topology implications.
67	2026-07	2607.26054	29	11	90.95	91.05	Uniqueness of shrinking Kähler-Ricci solitons on resolutions of Kähler cones Ronan J. Conlon; Alix Deruelle	Uniqueness theorem for shrinking KRS in singular/conical resolution setting.
68	2026-07	2607.27629	90	11	90.95	91.05	Curved Kaakeya problems and the projective geometry of paths Shaoming Guo; Larry Guth; Arian Nadjimiadzah; Minking Shen; Ruixiang Zhang	Highly technical incidence/harmonic-analysis result with projective-geometric content.
69	2026-07	2607.29447	-	11	90.95	91.05	The Penrose inequality with charge for 2-convex initial data sets Tuan Dolmen	Charged Penrose inequality under 2-convexity; strong GR geometry signal.
70	2026-07	2607.29588	-	11	90.95	91.05	A two-dimensional Allen-Cahn theory for interfaces with boundary Marco Badran; Manuel del Pino; Marco A. M. Guaraco	Boundary-interface Allen-Cahn theory with geometric variational content.
71	2026-08	2608.00944	56	11	90.95	91.05	McKean Rigidity for Cocompact Negatively Curved Manifolds and the p -Laplacian Kuntao Jin; Bo Zhu	Rigidity theorem for negatively curved manifolds and p -Laplacian; high spectral-geometric value.
72	2026-08	2608.03405	60	11	90.95	91.05	Lower Bound for Weighted Intermediate Ricci Curvature and Tensorial Entropy Convexity Chao-Ming Lin; Kai-Hsiang Wang	Weighted curvature lower bounds and entropy convexity; strong synthetic/geometric analysis link.
73	2026-08	2608.04364	9	11	90.95	91.05	Calabi-Yau Conjecture for Minimal Hypersurfaces in R^4 with bounded geometry Shrey Aryan; Alexander D. McWeeney; Giuseppe Tinaglia	Concise theorem around Calabi-Yau conjecture for minimal hypersurfaces.
74	2026-06	2606.09017	-	13	90.45	90.92	The Ollivier Ricci flow with prescribed curvature on infinite graphs Bobo Hua; Yong Lin; Shuang Liu	Prescribed-curvature Ollivier Ricci flow on infinite graphs.
75	2026-06	2606.28767	31	12	90.60	90.92	Infinite-Time Singularities with Vanishing Mean Curvature for Lagrangian Mean Curvature Flow in Gibbons-Hawking Spaces Ping-Hung Lee; Chung-Jun Tsai	Infinite-time singularities in Gibbons-Hawking Lagrangian MCF setting.
76	2026-06	2606.29231	10	12	90.70	90.85	On questions of Pogorelov and Toponogov Lei Ni; Wei Zhang; Yijian Zhang	Addresses classical Pogorelov-Toponogov type questions.
77	2026-06	2606.29246	31	12	90.70	90.85	Rigidity of Closed Minimal Hypersurfaces in S^5 Jianquan Ge; Tong Liu; Keyan Luo; Wenjiao Yan	Rigidity result for closed minimal hypersurfaces in the five-sphere.
78	2026-08	2608.02405	16	11	90.35	90.78	A Weighted Discretization of Riemannian Manifolds with Lower Ricci Bounds Aditya Tiwari	Weighted combinatorial Laplacian with spectral comparison independent of injectivity radius and stability under measured Gromov-Hausdorff convergence.
79	2026-08	2608.06974	7	12	90.30	90.72	Left-invariant metrics on step-two Carnot groups are uniformly doubling Cheng Bi; Ye Zhang	Proves the sharp uniform doubling constant $2^{**}Q$ for all left-invariant Riemannian metrics on every step-two Carnot group, together with a Bishop-Gromov-type estimate.
80	2026-07	2607.29654	23	12	90.05	90.50	Elastic Curves via Geometric Mechanics Oliver Gross; Rohit Jammula; Albert Chern	New isoperimetric characterization and exact structure-preserving polygonal theory; carries canonical symplectic dynamics to discrete curves.
81	2026-08	2507.16794	23	15	89.85	90.40	Spectral gaps for noncompact hyperbolic surfaces with linearly many cusps Qi Guo; Bobo Hua; Yang Shen	Closes the linear-cusp existence regime through a quantitative expander-graph construction and proves a complementary Steklov obstruction for superlinear cusp growth.
82	2026-07	2607.05023	45	11	90.35	90.40	Rigidity on compact surfaces through hyperbolic symmetries Sean Dewar; Alison La Porta; Rebecca Monks; Anthony Nixon; Klara Stokes; Joannes Vermant	First finite combinatorial characterization of generic rigidity for frameworks on compact surfaces of genus at least two, via hyperbolic symmetries and gain graphs.
83	2026-06	2606.31393	27	12	90.20	90.35	Montel's theorem and tautness in calibrated geometry Anton Iliashenko; Spiro Karigiannis; Jesse Madnick	Montel/tautness theory in calibrated geometry.
84	2026-06	2606.15402	43	13	89.90	90.30	Rigidity Theorems for the Weyl Problem of Convex Surfaces in Hyperbolic 3-Space Xinrong Zhao	Rigidity theorems for hyperbolic Weyl problem.
85	2026-06	2606.15827	-	13	89.90	90.30	Rigidity Estimate for Hyperbolic Shells and its Application in Gamma-limit Theory Liang-Biao Chen; Peng-Fei Yao	Rigidity estimate for hyperbolic shells with Gamma-limit applications.
86	2026-08	2608.07351	11	14	89.85	90.17	Finite-Rank Lie Algebroids for Singular Foliations of Prescribed Vanishing Order Austin Davis; Cheng-Yi Hung; Sangjun Ko; Kaichuan Qi	Solves a finite-rank realizability problem for prescribed-order vanishing singular foliations via a new construction from left-symmetric and sub-adjacent Lie algebroids.
87	2026-07	2607.02244	-	12	90.10	90.17	Foliations by constant spacetime mean curvature surfaces for asymptotically hyperboloidal initial data sets Jacopo Tenan	Foliation result in asymptotically hyperboloidal GR geometry.
88	2026-07	2607.05755	29	12	90.10	90.17	Topological Bernstein Theorems for Minimal Hypersurfaces in R^4 confined in space Shrey Aryan; Alexander D. McWeeney	Bernstein-type statement in R^4 under confinement assumptions.
89	2026-07	2607.05980	32	12	90.10	90.17	Charged parallel spinors and applications to mass-charge inequalities Simon Raulot	Spinorial approach to mass-charge inequalities.
90	2026-07	2607.06951	33	12	90.10	90.17	A sharp isoperimetric inequality and the top order Q-curvature Mingxiang Li; Xingwang Xu	Sharp inequality linked to top-order Q-curvature.
91	2026-07	2607.08622	48	12	90.10	90.17	Deformed Hermitian-Yang-Mills equation on the manifold of full flags Benoit Charbonneau; Gonçalo Oliveira; Rosa Sena-Dias	dHYM equation in a homogeneous complex-geometric setting.
92	2026-07	2607.10733	43	12	90.10	90.17	On Chern's Conjecture for Minimal Submanifolds with Flat Normal Bundle in Spheres Jianquan Ge; Fagui Li; Yunheng Zhang	Progress around Chern-type rigidity for minimal submanifolds.
93	2026-07	2607.11067	-	12	90.10	90.17	Blow-up phenomena for the sigma_2-Yamabe equation Bin Deng; Han Lu; Jun-cheng Wei	Blow-up analysis for fully nonlinear conformal equation.
94	2026-07	2607.12842	161	12	90.10	90.17	Lectures on Alexandrov spaces with curvature bounded below Vitali Kapovitch; Nina Lebedeva; Anton Petrunin	Major expository/reference value; originality discounted relative to research papers.
95	2026-07	2607.12979	31	12	90.10	90.17	Ambient geometry via min-max widths of embedded circles Lucas Ambrozio; Ivan Miranda; Rafael Montezuma	Uses min-max widths of curves to probe ambient geometry.
96	2026-07	2607.13450	63	12	90.10	90.17	Nonlinear Hodge correspondence for morphisms Nianzi Li; Mao Sheng	Hodge-correspondence result with significant geometric structure.
97	2026-07	2607.13742	30	12	90.10	90.17	A sharp relative comparison inequality for conformal fillings of Poincare-Einstein manifolds Nan Wu	Sharp comparison inequality in conformal/geometric analysis.
98	2026-07	2607.13760	31	12	90.10	90.17	Strict concavity of the growth indicator function for relatively Anosov groups Dongryul M. Kim; Hee Oh; Andrew Zimmer	Growth indicator concavity for relatively Anosov groups; strong dynamics/geometry cross-value.
99	2026-07	2607.13890	84	12	90.10	90.17	Heat Kernel and Closed Geodesic Asymptotics for Nilpotent Coverings Atsushi Katsuda	Heat-kernel and closed-geodesic asymptotics; useful spectral-geometric contribution.
100	2026-07	2607.14244	11	12	90.10	90.17	Scalar Curvature and Product Manifolds Claude LeBrun	Concise scalar-curvature result by a leading author; high conceptual signal.

June 2026 Monthly Ranking - Top 50

Official entries: 387; scored pool: 117. The 6-8 Rank column gives each paper's position in the complete combined comparison set.

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2	2	2606.03583	58	8	94.90	95.07	Closed minimal surfaces of index one in Riemannian manifolds Fernando C. Marques; André Neves	Index-one minimal surface theory by leading min-max authors; high conceptual and technical weight.
3	3	2606.12257	333	8	94.70	94.85	Quantum cohomology and split generation in Lagrangian Floer theory Mohammed Abouzaid; Kenji Fukaya; Yong-Geun Oh; Hiroshi Ohta; Kaoru Ono	Massive foundational Floer-theoretic contribution with strong downstream value.
4	4	2606.30098	77	8	94.05	94.38	Min-Max Construction of Anisotropic Minimal Surfaces with Genus Bound Antonio De Rosa; Aria Halavati; Ling Wang	Min-max construction with genus control in anisotropic surface theory.
5	7	2606.13335	59	8	93.75	94.07	Sharp stability of Alexandrov's theorem for C^1 domains in the small-excess regime Alessio Figalli; Yi Ru-Ya Zhang	Sharp quantitative stability for Alexandrov-type rigidity.
6	8	2606.15795	44	8	93.75	94.07	Entire area-minimizing surfaces in R^n of density two are quadratic or planar Nick Edelen; Luis Atzin Franco Reyna; Paul Minter	Sharp classification for density-two entire area minimizers.
7	9	2606.22355	-	8	93.75	94.00	Uncountably many non-rotationally symmetric type II ancient Yamabe flows on the sphere Haixia Chen; Seunghyeok Kim; Monica Musso	Constructs large family of type-II ancient Yamabe flows with symmetry breaking.
8	11	2606.00964	56	9	93.40	93.55	Mean convex flows with surgery Gabor Szekelyhidi	Mean-convex MCF with surgery; high technical geometric-flow relevance.
9	12	2606.15816	-	9	93.40	93.55	Kähler-Ricci Flow: from divisors to cusps Eleonora Di Nezza; Vincent Guedj; Chinh H. Lu	Kähler-Ricci flow transition from divisorial to cusp singularities.
10	14	2606.30130	74	9	93.05	93.38	Gromov's dihedral rigidity conjecture in dimension three Jinmin Wang; Zhizhang Xie; Guoliang Yu	Major scalar-curvature rigidity conjecture in dimension three.
11	19	2606.29135	27	9	93.05	93.15	Gromov's Simplicial Volume Vanishing Conjecture for Positive Scalar Curvature and Fundamental Group Decay Qiaochu Ma; Guoliang Yu	Important PSC-topology result toward Gromov-type vanishing conjectures.
12	20	2606.21062	-	9	92.75	93.07	The prescribed Hermitian-Yang-Mills flow I Zhiyao Xiong; Xiaokui Yang; Shing-Tung Yau	First part of prescribed HYM flow program; strong complex differential geometry content.
13	21	2606.21073	-	9	92.75	93.07	The prescribed Hermitian-Yang-Mills flow II Zhiyao Xiong; Xiaokui Yang; Shing-Tung Yau	Second part of prescribed HYM flow program; complements Part I.
14	24	2606.13301	-	9	92.55	92.62	Gromov-Hausdorff convergence of time-slices of singular Ricci flows in dimension three Yu Li	GH convergence analysis for singular 3D Ricci flows.
15	25	2606.21541	64	9	92.55	92.62	Spectral Analysis for Finite-Time Singularities of Lagrangian Mean Curvature Flow Maxwell Stolarski; Wei-Bo Su	Spectral analysis around finite-time singularity models in Lagrangian MCF.
16	35	2606.02396	22	10	92.05	92.38	Existence of free boundary minimal disks in convex regions Lorenzo Santataro; Douglas Stryker; Zhichao Wang; Xin Zhou	Existence result for free-boundary minimal disks in convex regions.
17	36	2606.24682	-	10	92.05	92.38	The linearized minimal surfaces problem Haim Grebnev; Plamen Stefanov; Gunther Uhlmann; Hanming Zhou	Linearized inverse-type problem for minimal surfaces with strong analysis component.
18	37	2606.24754	-	10	92.05	92.30	Existence of classical minimal surfaces in 4 and 5-manifolds Da Rong Cheng; Xin Zhou	Classical minimal surface existence in low-dimensional manifolds.
19	38	2606.13011	71	10	91.75	92.08	Complete Calabi-Yau metrics on noncompact K3 fibered threefolds Ruiming Liang	Construction and analysis of complete CY metrics on noncompact K3-fibered threefolds.
20	39	2606.16033	-	10	91.75	92.08	Admissibility, Topology, and the sigma_k-Loewner-Nirenberg problem Hao Fang; Matthew J. Gursky	Topology and admissibility in the sigma_k-Loewner-Nirenberg problem.
21	40	2606.22441	22	10	91.75	92.08	Parabolic rectifiability of the Brakke flow Yu Tong Liu; Myles Workman	Parabolic rectifiability for Brakke flows; useful geometric measure theory tool.
22	42	2606.04586	43	10	91.75	91.85	Calibration energy and mean curvature flow Tatsuya Miura; Fabian Rupp	Calibration-energy perspective for MCF; technically robust.
23	43	2606.08006	23	12	91.55	91.78	Optimal spectral rigidity of the hypercube via Bakry-Emery curvature Yanlong Ding; Shiping Liu; Chiyu Zhou	Sharp discrete curvature/spectral rigidity on the hypercube.
24	45	2606.08086	28	11	91.05	91.38	Monotone quantities on 3-manifolds with nonnegative scalar curvature Jiannan Chen; Haiping Fu	Monotone quantities in 3D nonnegative scalar curvature geometry.
25	46	2606.12063	-	11	90.75	91.07	Optimal geometric estimates for compact Kähler manifolds of a Nash entropy bound Weiqi Zhang; Yashan Zhang	Optimal estimates under Nash entropy bounds in compact Kähler setting.
26	47	2606.12379	21	11	90.75	91.07	A Local Singularity Analysis for the Ricci Flow and its Applications to Ricci Flows with Bounded Scalar Curvature -- Part II Reto Buzano; Gianmichele Di Matteo	Local singularity analysis for scalar-curvature-bounded Ricci flows.
27	48	2606.19801	-	11	90.75	91.07	Positive mass theorem and the Yamabe equation on CR manifolds Jih-Hsin Cheng	Positive mass theorem linked to CR Yamabe equation.
28	49	2606.19806	10	11	90.75	91.07	The top Yau-Yang conjecture for Kähler manifolds with positive sectional curvature Ved V. Datar; Vamsi P. Pingali; Harish Seshadri	Concise result on the top Yau-Yang conjecture for positively curved Kähler manifolds.
29	50	2606.19858	16	11	90.75	91.07	Spectral Positive Mass Theorem for Asymptotically Hyperbolic 3-manifolds with Toroidal Infinity Xiaoxiang Chai; Yimin Chen; Juncheol Pyo	Spectral positive mass theorem in AH three-manifolds with toroidal infinity.
30	51	2606.28897	16	12	90.90	91.07	Scalar-Flatness for Critical Metrics of the L2-Scalar Curvature Functional in Dimensions 5<=n<=9 Heng Zhang	Scalar-flatness for critical L2-scalar curvature metrics in middle dimensions.
31	74	2606.09017	-	13	90.45	90.92	The Ollivier Ricci flow with prescribed curvature on infinite graphs Bobo Hua; Yong Lin; Shuang Liu	Prescribed-curvature Ollivier Ricci flow on infinite graphs.
32	75	2606.28767	31	12	90.60	90.92	Infinite-Time Singularities with Vanishing Mean Curvature for Lagrangian Mean Curvature Flow in Gibbons-Hawking Spaces Ping-Hung Lee; Chung-Jun Tsai	Infinite-time singularities in Gibbons-Hawking Lagrangian MCF setting.
33	76	2606.29231	10	12	90.70	90.85	On questions of Pogorelov and Toponogov Lei Ni; Wei Zhang; Yijian Zhang	Addresses classical Pogorelov-Toponogov type questions.
34	77	2606.29246	31	12	90.70	90.85	Rigidity of Closed Minimal Hypersurfaces in S^5 Jianquan Ge; Tong Liu; Keyan Luo; Wenjiao Yan	Rigidity result for closed minimal hypersurfaces in the five-sphere.
35	83	2606.31393	27	12	90.20	90.35	Montel's theorem and tautness in calibrated geometry Anton Ilashenko; Spiro Karigiannis; Jesse Madnick	Montel/tautness theory in calibrated geometry.
36	84	2606.15402	43	13	89.90	90.30	Rigidity Theorems for the Weyl Problem of Convex Surfaces in Hyperbolic 3-Space Xinrong Zhao	Rigidity theorems for hyperbolic Weyl problem.
37	85	2606.15827	-	13	89.90	90.30	Rigidity Estimate for Hyperbolic Shells and its Application in Gamma-limit Theory Liang-Biao Chen; Peng-Fei Yao	Rigidity estimate for hyperbolic shells with Gamma-limit applications.
38	117	2606.08648	20	12	89.75	90.07	On Brezis Open Problem 3.1 Qi Guo; Xueping Huang; Yi C. Huang; Fanghua Lin; Juncheng Wei	Resolves/advances Brezis Open Problem 3.1 in a geometric-analytic setting.
39	118	2606.22468	59	12	89.75	90.07	A Classical Elliptic Regularity Approach to Almost Harmonic Maps and Related Systems Giovanni Di Fratta	Elliptic regularity framework for almost harmonic maps and related systems.
40	119	2606.22777	41	12	89.75	90.07	Weak Quadruple Comparison and Structure Theory Beyond Alexandrov Geometry Bang-Xian Han; Liming Yin	Structure theory beyond Alexandrov geometry via weak quadruple comparison.
41	120	2606.14513	-	13	89.60	90.00	Quasilinear Liouville equation on manifolds with nonnegative Ricci curvature Giovanni Catino; Dario Daniele Monticelli; Alberto Roncoroni	Liouville-type theorem under nonnegative Ricci curvature.
42	121	2606.15599	67	13	89.60	90.00	Non-unital monoidal category of contact manifolds and Legendrian correspondence Yong-Geun Oh; Junhyuk Park	Categorical contact/Legendrian framework with high conceptual content.
43	122	2606.27155	22	13	89.60	89.92	Riemannian Penrose Inequality for Manifolds with Corners via Non-Linear Potential Theory Virginia Agostiniani; Christina Karapiperi; Lorenzo Mazziere	Riemannian Penrose inequality with corners using nonlinear potential theory.
44	123	2606.14021	-	13	89.60	89.78	Some Foundational Results for Free Boundary Brakke Flows Yueheng Bao	Foundational free-boundary Brakke flow results.
45	125	2606.14082	-	13	89.40	89.55	Inverse Hessian Curvature Flow in Minkowski Space I: Cocompact Hypersurfaces Dake Li; Zhizhang Wang; Shiqi Yin	Inverse Hessian curvature flow in Minkowski space.
46	126	2606.14318	-	13	89.40	89.55	Sectional Curvature for Kantorovich-Wasserstein and Hellinger-Kantorovich Geometries Karl-Theodor Sturm	Sectional curvature notions for Wasserstein and Hellinger-Kantorovich geometries.
47	129	2606.01153	34	14	88.90	89.30	Scalar Curvature, Volumes and the Bergman Kernel Bo-Yong Chen; Yuanpu Xiong; Liyou Zhang	Links scalar curvature, volume, and Bergman kernel estimates.
48	145	2606.10443	-	18	88.65	89.17	Geometry of gravitational instantons Mingyang Li; Song Sun	High-level survey on gravitational instanton geometry.
49	148	2606.00354	24	14	88.60	89.00	Inequivalence of the various notions of completeness for Cartan geometries Jacob W. Erickson; Benjamin McKay	Clarifies inequivalent completeness notions in Cartan geometry.
50	149	2606.01108	-	14	88.60	89.00	Integral type Gauss-Green formula on non-collapsed RCD spaces and its applications Zhangkai Huang	Gauss-Green formula on non-collapsed RCD spaces.

July 2026 Monthly Ranking - Top 50

Official entries: 331; scored pool: 100. The 6-8 Rank column gives each paper's position in the complete combined comparison set.

M.Rank	6-8 Rank	arXiv	Pages	AI%	S_H	S_eq	Title / Authors	Assessment
1	6	2607.13356	156	8	94.00	94.35	Analysis of singularities of area-minimizing currents, Part III: branch points of planar frequency != 2, higher order asymptotics, and the local topology Brian Krummel; Neshan Wickramasekera	Large-scale regularity/singularity analysis for area-minimizing currents; exceptional human proof complexity.
2	5	2607.07134	116	8	94.00	94.35	Naked Singularities beyond Spherical Symmetry: Singular Inner Cauchy Horizons for the Einstein-Scalar Field System Xinliang An; Shengrong Wu	Major GR-PDE construction beyond spherical symmetry; strong geometric analysis relevance.
3	10	2607.15522	34	9	93.00	93.65	Nonnegative Bakry-Émery Curvature on Bounded-Degree Graphs Implies Volume Doubling and Poincaré Inequalities Qi Guo; Xueping Huang; Yi C. Huang	Bounded-degree CD(0,∞) graphs are uniformly volume doubling and satisfy a scale-invariant L2 Poincaré inequality; settles the Cushing-Liu-Peyerimhoff polynomial-growth conjecture in a stronger form.
4	17	2607.11236	-	9	93.15	93.33	Formation of Trapped Surfaces from Spacelike Initial Data Xuantao Chen; Sergiu Klainerman	High-value trapped-surface formation result; long nonlinear geometric PDE proof architecture.
5	15	2607.03152	110	9	93.15	93.33	Finite-Time Singularities of Lagrangian Mean Curvature Flow with Quantitatively Precise Dynamics Maxwell Stolarski; Wei-Bo Su	Quantitative singularity dynamics for Lagrangian MCF; technically deep and reusable.
6	16	2607.03383	78	9	93.15	93.33	Unique asymptotics of SO(k)xSO(n-k+1) symmetric ancient ovals of Ricci flow Panagiota Daskalopoulos; Wenkui Du; Natasa Sesum; Ziyi Zhao	Ancient Ricci-flow asymptotics by leading flow techniques; high geometric PDE value.
7	32	2607.29502	-	10	92.30	92.45	A dichotomy for minimal submanifolds Tobias Holck Colding; William P. Minicozzi II	Conceptual dichotomy statement in minimal-submanifold theory by leading authors.
8	33	2607.29632	60	10	92.30	92.45	Regularity of branched stable minimal immersed hypersurfaces Gaoming Wang; Xuwen Zhang	Regularity theory for branched stable minimal hypersurfaces; substantial proof burden.
9	27	2607.03930	-	10	92.30	92.45	De Giorgi varifold solutions to Mean Curvature Flow: a minimizing movements approach Tim Laux; Andrea Poiatti	Variational weak solution framework for MCF; high method transfer potential.
10	31	2607.25644	62	10	92.30	92.45	Gromov-Hausdorff Limits of Noncollapsed Kähler-Ricci Flows and the Geometry of Ricci Shrinkers Yu Li; Junsheng Zhang	Noncollapsed KRF limits and shrinker geometry; central geometric-flow problem area.
11	28	2607.06688	-	10	92.30	92.45	Poisson K-stability and the semiclassical Yau-Tian-Donaldson correspondence Vestislav Apostolov; Brent Pym; Jeffrey Streets	Connects K-stability/YTD ideas with Poisson geometry; high conceptual breadth.
12	30	2607.23879	40	10	92.30	92.45	Closed mean curvature flows with prescribed tangent flows Jingwen Chen; Tang-Kai Lee; Ao Sun; Xinrui Zhao	Prescribed tangent-flow phenomena for closed MCF; strong singularity-theory relevance.
13	29	2607.20741	-	10	92.30	92.45	The Penrose conjecture for initial data sets satisfying a 2-convexity condition Conghan Dong	Penrose-type result under geometric convexity; important in scalar-curvature/GR interface.
14	41	2607.27162	23	10	91.65	91.88	Pointwise subexponential growth and near-diffusive displacement on bounded-degree graphs with non-negative Ollivier-Ricci curvature Chiyu Zhou	Near-diffusive displacement and pointwise subexponential volume growth for bounded-degree graphs with nonnegative Ollivier-Ricci curvature.
15	58	2607.17459	88	11	90.95	91.05	The deformed Vortex equations and equivariant stability conditions Tristan C. Collins; Yukai Zhang	Gauge/Kähler stability problem with substantial length and technical structure.
16	70	2607.29588	-	11	90.95	91.05	A two-dimensional Allen-Cahn theory for interfaces with boundary Marco Badran; Manuel del Pino; Marco A. M. Guaraco	Boundary-interface Allen-Cahn theory with geometric variational content.
17	66	2607.25015	36	11	90.95	91.05	Topological Rigidity of Contractible 3-Manifolds and Handlebody Interiors under Nonnegative Scalar Curvature Jiangcheng You; Heng Zhang	Scalar-curvature rigidity for contractible 3-manifolds; strong geometric topology implications.
18	67	2607.26054	29	11	90.95	91.05	Uniqueness of shrinking Kähler-Ricci solitons on resolutions of Kähler cones Ronan J. Conlon; Alix Deruelle	Uniqueness theorem for shrinking KRS in singular/conical resolution setting.
19	55	2607.08325	30	11	90.95	91.05	Geometric smoothing by the Kähler-Ricci Flow Eleonora Di Nezza; Vincent Guedj; Chinh Hoang Lu	KRF smoothing in complex geometric analysis; useful technical input.
20	56	2607.08631	32	11	90.95	91.05	Existence of two embedded minimal spheres in S^~3 with an arbitrary metric Zhichao Wang; Xin Zhou	Existence result for embedded minimal spheres; strong min-max/geometric analysis value.
21	57	2607.12305	64	11	90.95	91.05	Capillary prisms, Coxeter gluing, and the pi_2-systole of 3-manifolds with positive scalar curvature Shihang He; Chao Li	Positive scalar curvature, capillary geometry and systolic estimates; broad geometric interest.
22	54	2607.06698	151	11	90.95	91.05	Infinity-harmonic functions and inverse mean curvature flow clusters Kai Xu	Very long analysis of infinity-harmonic and IMCF cluster structures.
23	53	2607.02412	-	11	90.95	91.05	Cartan's and Gauss's equations and rigidity theorems for isometric embeddings in low Sobolev regularity Isaac Newell; Luc Nguyen	Low-Sobolev rigidity for isometric embeddings; high technical and analytic value.
24	52	2607.02067	31	11	90.95	91.05	Higher regularity of the inverse anisotropic mean curvature flow Chaoqun Gao; Yong Wei; Rong Zhou	Regularity upgrade for inverse anisotropic MCF; solid geometric flow result.
25	64	2607.24281	45	11	90.95	91.05	Calabi surgery for Z/Z harmonic 1-forms Jiahuang Chen; Siqi He; Dashen Yan	Surgery construction for Z/Z harmonic 1-forms; relevant to gauge/geometric topology.
26	60	2607.18792	35	11	90.95	91.05	Existence of global W^~{2,p}-isometric immersions for negatively curved surface metrics with unbounded second fundamental form Siran Li	Global low-regularity isometric immersion under negative curvature; strong PDE-geometry link.
27	61	2607.18989	103	11	90.95	91.05	Diffeological non-Abelian Hodge theory: relative harmonic metrics and deformation theory Mahmud Azam; Steven Rayan	Large deformation/Hodge theory work with differential-geometric analytic component.
28	62	2607.19091	32	11	90.95	91.05	On the sharp constants in curl-Sobolev inequalities on S^~n Guofang Wang; Mingwei Zhang	Sharp inequality work on spheres; relevant across DG, AP and spectral theory.
29	63	2607.19320	-	11	90.95	91.05	A spacetime positive mass theorem with corners via mollification Hyun Chul Jang; Aidan F. Wood	Positive mass theorem with corners; important low-regularity GR geometry theme.
30	65	2607.24735	-	11	90.95	91.05	Area-minimizing submanifolds are not generically smooth, except for geodesics, minimal surfaces, and minimal hypersurfaces Zhenhua Liu	Striking generic regularity/nonregularity statement for area minimizers.
31	68	2607.27629	90	11	90.95	91.05	Curved Kakeya problems and the projective geometry of paths Shaoming Guo; Larry Guth; Arian Nadjmi-zadah; Minxing Shen; Ruixiang Zhang	Highly technical incidence/harmonic-analysis result with projective-geometric content.
32	59	2607.18216	-	11	90.95	91.05	Sharp Weitzenböck and PIC2 Estimates from Sectional-Scalar Curvature Pinching Jian Ge	Sharp curvature estimates from mixed sectional-scalar pinching; strong comparison geometry.
33	69	2607.29447	-	11	90.95	91.05	The Penrose inequality with charge for 2-convex initial data sets Tuan Dolmen	Charged Penrose inequality under 2-convexity; strong GR geometry signal.
34	80	2607.29654	23	12	90.05	90.50	Elastic Curves via Geometric Mechanics Oliver Gross; Rohit Jammula; Albert Chern	New isoperimetric characterization and exact structure-preserving polygonal theory; carries canonical symplectic dynamics to discrete curves.
35	82	2607.05023	45	11	90.35	90.40	Rigidity on compact surfaces through hyperbolic symmetries Sean Dewar; Alison La Porta; Rebecca Monks; Anthony Nixon; Klara Stokes; Joannes Vermant	First finite combinatorial characterization of generic rigidity for frameworks on compact surfaces of genus at least two, via hyperbolic symmetries and gain graphs.
36	105	2607.20350	38	12	90.10	90.17	Modified Mean Curvature Flow in Fuchsian Manifolds Yuk Shing Lam; Longzhi Lin	Flow theory in Fuchsian manifolds; solid specialized geometric PDE contribution.
37	106	2607.21031	39	12	90.10	90.17	Structure theory of BV functions and finite perimeter sets on Riemannian manifolds Péter Koltai; Kathrin Völkner	Metric/geometric measure-theoretic structure theory on manifolds.
38	107	2607.23703	42	12	90.10	90.17	Plateau's Problem via covering spaces James Tisot	Plateau problem reformulation via covering spaces; broad classical problem context.
39	109	2607.23872	-	12	90.10	90.17	A Partial epsilon-Regularity Theorem for Integral Stationary Geodesic Networks in R^~n Henry Bosch	Partial regularity for stationary geodesic networks; technical GMT contribution.
40	108	2607.23827	-	12	90.10	90.17	The sharp curl-Sobolev inequality Guofang Wang; Mingwei Zhang	Sharp Sobolev-type inequality; shorter but with useful analytic geometry impact.
41	110	2607.26302	34	12	90.10	90.17	Special Lagrangians with multiple isolated singularities Bryan Dimler; Filippo Gaia	Special Lagrangian singularity construction; good calibrated-geometry relevance.
42	113	2607.29667	20	12	90.10	90.17	The large mass limit of monopoles: abelian limits and Dirac singularities Daniel Fadel	Gauge-theoretic asymptotic limit with Dirac singularities; mathematically focused.
43	99	2607.13890	84	12	90.10	90.17	Heat Kernel and Closed Geodesic Asymptotics for Nilpotent Coverings Atsushi Katsuda	Heat-kernel and closed-geodesic asymptotics; useful spectral-geometric contribution.
44	102	2607.15106	26	12	90.10	90.17	A quantitative Schur comparison theorem for curves in CAT(k) spaces Mohammad Ghomi; John Ioannis Stavroulakis	Metric comparison theorem with quantitative content.
45	97	2607.13742	30	12	90.10	90.17	A sharp relative comparison inequality for conformal fillings of Poincare-Einstein manifolds Nan Wu	Sharp comparison inequality in conformal/geometric analysis.
46	98	2607.13760	31	12	90.10	90.17	Strict concavity of the growth indicator function for relatively Anosov groups Dongryul M. Kim; Hee Oh; Andrew Zimmer	Growth indicator concavity for relatively Anosov groups; strong dynamics/geometry cross-value.
47	100	2607.14244	11	12	90.10	90.17	Scalar Curvature and Product Manifolds Claude LeBrun	Concise scalar-curvature result by a leading author; high conceptual signal.
48	101	2607.14857	-	12	90.10	90.17	A framework for producing harmonic maps Volker Branding; Anna Siffert	General harmonic-map production framework; solid but moderate novelty signal.
49	94	2607.12842	161	12	90.10	90.17	Lectures on Alexandrov spaces with curvature bounded below Vitali Kapovitch; Nina Lebedeva; Anton Petrunin	Major expository/reference value; originality discounted relative to research papers.
50	95	2607.12979	31	12	90.10	90.17	Ambient geometry via min-max widths of embedded circles Lucas Ambrozio; Ivan Miranda; Rafael Montezuma	Uses min-max widths of curves to probe ambient geometry.

August 2026 Monthly Ranking - All 21 Papers in the Official 30% Pool

Official entries through 10 August: 70; scored pool: 21. The monthly table retains the official new/cross-list selection.

M.Ra...	6-8 Rank	arXiv	Pages	AI%	S_H	S_eq	Title / Authors	Assessment
1	13	2608.07240	59	10	92.90	93.45	Critical Geller Equations on Complex Hyperbolic Space: Nondegeneracy, Geodesic Symmetry, and Global Compactness Jungang Li	Builds a broad critical Geller/Brezis-Nirenberg theory: sharp constants and classification, invariant-sector profiles and stability, nondegeneracy,....
2	18	2608.05128	148	9	93.15	93.32	Large mass limits of G_2 and Calabi-Yau monopoles: calibrated concentration, Higgs zeros, and abelianization Daniel Fadel; Gonalo Oliveira	Very large gauge/calibrated-geometry work; high human proof burden and structural value.
3	22	2608.06320	18	8	92.45	93.00	Dihedral Rigidity for Convex Polytopes by Smooth Approximation Yuchen Bi	Smooth-approximation proof of Gromov's dihedral rigidity conjecture for convex polytopes.
4	23	2608.07231	25	10	92.50	92.72	Liouville Rigidity for Entire Strictly Spacelike Solutions of a Lorentzian Prescribed Mean Curvature Equation Xi-Nan Ma; Tian Wu; Wangzhe Wu; Bao Yu	Removes radial, decay, integrability, and uniform-gap assumptions in a subcritical Lorentzian prescribed-mean-curvature Liouville problem, using...
5	26	2608.03575	39	10	92.10	92.50	A Gauss-Bonnet-Type Dichotomy for Unimodular Random Infinite Trivalent Hyperbolic Polyhedra Huabin Ge; Yangxiang Lu; Chuwen Wang; Tian Zhou	Unimodular Gauss-Bonnet dichotomy, finite-limit obstruction, a refined ring lemma, Poisson-boundary identification, and positive hyperbolic speed.
6	34	2608.01682	42	10	92.30	92.45	Positive mass theorems for singular asymptotically hyperbolic manifolds Yuguang Shi; Chengzhang Sun; Zijun Wang	Positive mass theorem in singular asymptotically hyperbolic setting; strong geometric analysis.
7	44	2608.07263	16	12	91.20	91.50	Diagonalization of the Toda flow for arbitrary isospectral symmetric matrices David Martnez Torres; Carlos Tomei	Constructs a dense covering atlas that linearizes the Toda flow on arbitrary symmetric isospectral manifolds, based on a new constrained QLU...
8	71	2608.00944	56	11	90.95	91.05	McKean Rigidity for Cocompact Negatively Curved Manifolds and the p-Laplacian Kuntao Jin; Bo Zhu	Rigidity theorem for negatively curved manifolds and p-Laplacian; high spectral-geometric value.
9	72	2608.03405	60	11	90.95	91.05	Lower Bound for Weighted Intermediate Ricci Curvature and Tensorial Entropy Convexity Chao-Ming Lin; Kai-Hsiang Wang	Weighted curvature lower bounds and entropy convexity; strong synthetic/geometric analysis link.
10	73	2608.04364	9	11	90.95	91.05	Calabi-Yau Conjecture for Minimal Hypersurfaces in R^4 with bounded geometry Shrey Aryan; Alexander D. McWeeney; Giuseppe Tinaglia	Concise theorem around Calabi-Yau conjecture for minimal hypersurfaces.
11	78	2608.02405	16	11	90.35	90.78	A Weighted Discretization of Riemannian Manifolds with Lower Ricci Bounds Aditya Tiwari	Weighted combinatorial Laplacian with spectral comparison independent of injectivity radius and stability under measured Gromov-Hausdorff...
12	79	2608.06974	7	12	90.30	90.72	Left-invariant metrics on step-two Carnot groups are uniformly doubling Cheng Bi; Ye Zhang	Proves the sharp uniform doubling constant 2^Q for all left-invariant Riemannian metrics on every step-two Carnot group, together with a...
13	86	2608.07351	11	14	89.85	90.18	Finite-Rank Lie Algebroids for Singular Foliations of Prescribed Vanishing Order Austin Davis; Cheng-Yi Hung; Sangjun Ko; Kaichuan Qi	Solves a finite-rank realizability problem for prescribed-order vanishing singular foliations via a new construction from left-symmetric and...
14	114	2608.03442	-	12	90.10	90.17	Lipschitz regularity of harmonic map heat flows from RCD spaces into CAT(0) spaces Bang-Xian Han; Hui-Chun Zhang; Xi-Ping Zhu	Regularity of harmonic map heat flows in nonsmooth target/source geometry.
15	115	2608.03572	30	12	90.10	90.17	CscK metrics on birational models of projective varieties Zakarias Sjstrm Dyrefelt	cscK existence on birational models; central Khler geometry theme.
16	116	2608.04718	26	12	90.10	90.17	Ideal points, directed completion and the case of the maximally extended Schwarzschild spacetime Nicola Gigli; Argam Ohanyan; Marco Picerni; Zhe-Feng Xu; Matteo Zanardini	Metric completion viewpoint on Schwarzschild spacetime; good Lorentzian-geometry contribution.
17	124	2608.07021	16	14	89.00	89.55	Monte Carlo methods on compact symplectic manifolds Yuri A. Kordyukov	Transfers determinantal-point-process quadrature to prequantized compact symplectic manifolds and proves a central limit theorem with optimal...
18	130	2608.06678	37	15	89.05	89.20	Transverse stable causality in Lorentzian foliations Henrique Amador Puel Martins; Ivan Pontual Costa e Silva; Victor Luis Espinoza	Introduces transverse stable causality for Lorentzian foliations, proves several structural implications and near-equivalence for simple foliations; the full...
19	142	2608.02370	54	13	89.10	89.17	The Geometry and Dynamics of Spiral Minimal Products Haizhong Li; Yongsheng Zhang	Spiral minimal products; strengthens earlier results with geometric dynamics.
20	143	2608.02840	-	13	89.10	89.17	Homogeneous spaces with two equivalent isotropy summands develop positive Ricci curvature under Ricci flow Eric Cochran; Arseny Mingajev; Lawrence Mouill; Nazia Valiyakath	Ricci-flow positivity result for homogeneous spaces.
21	144	2608.03982	-	13	89.10	89.17	Analysis on surfaces with locally bounded integral curvature Sebastian Boldt; Batu Gneysu; Maxime Marot	Analysis framework on surfaces with bounded integral curvature.

10 August 2026 math.DG New/Cross-list Batch - All 11 Assessed

All papers in the official batch receive component scores. The combined rank is reported in the workbook.

D.R...	arXiv	Pages	AI%	V	S_H	S_eq	Title / Authors	Assessment
1	2608.07240	59	10	94.00	92.90	93.45	Critical Geller Equations on Complex Hyperbolic Space: Nondegeneracy, Geodesic Symmetry, and Global Compactness Jungang Li	Builds a broad critical Geller/Brezis-Nirenberg theory: sharp constants and classification, invariant-sector profiles and stability, nondegeneracy, compactness, symmetry, asymptotics, and...
2	2608.07231	25	10	92.95	92.50	92.72	Liouville Rigidity for Entire Strictly Spacelike Solutions of a Lorentzian Prescribed Mean Curvature Equation Xi-Nan Ma; Tian Wu; Wangzhe Wu; Bao Yu	Removes radial, decay, integrability, and uniform-gap assumptions in a subcritical Lorentzian prescribed-mean-curvature Liouville problem, using global Lorentz-factor bounds and new weighted tensor...
3	2608.07263	16	12	91.80	91.20	91.50	Diagonalization of the Toda flow for arbitrary isospectral symmetric matrices David Martínez Torres; Carlos Tomei	Constructs a dense covering atlas that linearizes the Toda flow on arbitrary symmetric isospectral manifolds, based on a new constrained QLU factorization of independent interest.
4	2608.06974	7	12	91.15	90.30	90.72	Left-invariant metrics on step-two Carnot groups are uniformly doubling Cheng Bi; Ye Zhang	Proves the sharp uniform doubling constant 2^Q for all left-invariant Riemannian metrics on every step-two Carnot group, together with a Bishop-Gromov-type estimate.
5	2608.07351	11	14	90.50	89.85	90.18	Finite-Rank Lie Algebroids for Singular Foliations of Prescribed Vanishing Order Austin Davis; Cheng-Yi Hung; Sangjun Ko; Kaichuan Qi	Solves a finite-rank realizability problem for prescribed-order vanishing singular foliations via a new construction from left-symmetric and sub-adjacent Lie algebroids.
6	2608.07021	16	14	90.10	89.00	89.55	Monte Carlo methods on compact symplectic manifolds Yuri A. Kordyukov	Transfers determinantal-point-process quadrature to prequantized compact symplectic manifolds and proves a central limit theorem with optimal worst-case mean-square-error decay.
7	2608.06678	37	15	89.35	89.05	89.20	Transverse stable causality in Lorentzian foliations Henrique Amador Puel Martins; Ivan Pontual Costa e Silva; Victor Luis Espinoza	Introduces transverse stable causality for Lorentzian foliations, proves several structural implications and near-equivalence for simple foliations; the full general equivalence remains open.
8	2608.06935	6	15	89.15	87.90	88.52	Rigidity for spin fill-ins with scalar curvature bounded from below Bernd Ammann; Samuel Lockman	Establishes equality rigidity in the hyperspherical-radius bound for spin fill-ins, characterizing hyperbolic geodesic balls; a concise extension built on strong spinorial inputs.
9	2608.07227	14	16	88.65	87.85	88.25	Quillen equivalences for truncated multicomplexes Jiahao Hu	Proves Quillen equivalences between successive truncations of multicomplexes and answers a model-category question relevant to comparisons between almost-complex and complex geometry.
10	2608.06645	17	17	87.10	86.80	86.95	Causal Invariance of the Contact Number of Pseudo-Riemannian Submanifolds Juan S. Gómez	Proves that ordinary, spacelike, and timelike contact numbers agree at every finite order and gives an explicit Lorentzian contact-six example; a specialized finite-order extension.
11	2608.06855	74	20	86.85	85.50	86.18	A Deep Dive Into the Tangent Category of Schemes Geoff Vooyo	A detailed, largely expository construction of the tangent category of schemes, with an original differential-bundle reconstruction theorem; originality is discounted because most of the paper is expository.